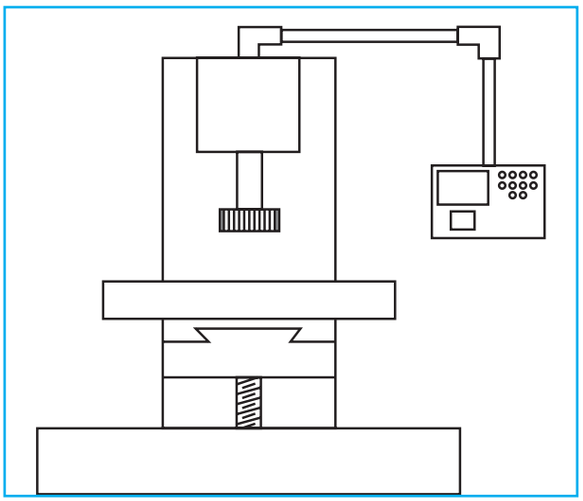
B. swivel MCU

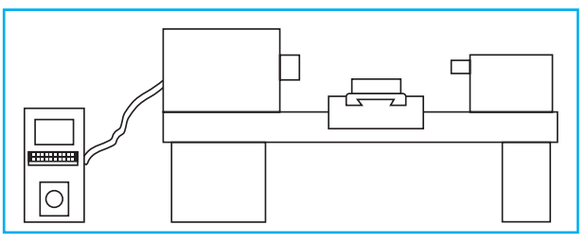
The MCU is a separate unit and is connected to the machine by swinging arrangements. The MCU as a whole can be swing around the machine depending upon the position of the operator. Swivel MCU is shown in figure.



Swivel MCU

C. Stand alone MCU

Stand alone MCU is designed as a separate unit and placed at a distance from the machine. Stand alone MCU is shown in figure.



Stand Alone MCU

Elements of a machine control unit

A machine control unit consists of following elements

1. Input reading unit
2. Memory unit
3. Processing unit
4. Output unit
5. Feedback unit

Input Reading Unit

It is a electromagnetic device useful in reading the NC programs in the form of instructions.

The main functions of an input reading unit are

1. To accept the input media
2. To collect the input media instructions.
3. To send them into processing unit.
4. To keep the input media safety for reuse.

Memory Unit

The information received from input reading unit will be saved in memory unit. It will be send to processing unit, when it is required.

Processing Unit

The informations received from the memory unit are processed here and are given to various parts of the CNC machine. The processing unit serves as a link between the memory unit and output unit.

Output Unit

The wires coming out of the processing unit are called output unit. This output unit emit electric pulses of low voltage. In order to drive various slides of the machines, the pulses are amplified by means of electronic or electro-magnetic amplifiers.

Feedback Unit

Feedback unit consists of some electrical or electronic hardware. This is used for converting physical quantities like displacement and velocity into electrical pulses.

XII BME_English version CHAPTER 09.indd 158 1/8/2020 1:38:49 PM